

Assignment 1: Floating Point Computation and Direct Linear System Solvers

October 24, 2025

Madeline LeBreton
MECHTRON 3X03

1. Suppose you approximate e^x by its truncated Maclaurin series. For $x = 0.5$, derive how many terms of the series are needed to achieve (absolute) truncation error of less than 10^{-10}

Ans: For any choice of real-numbered c and x in the range $[a, b]$ where the function is defined, the truncated series error is E_{n+1} :

$$E_{n+1} = \frac{f^{(n+1)}(z)}{(n+1)!} x^{n+1}$$

Let $z = x$, since e^x is an increasing function with its highest value at the rightmost point, where the error would be the largest.

$$E_{n+1} = \frac{e^x}{(n+1)!} (0.5)^{n+1}$$

$$10^{-10} \leq \frac{e^{0.5}}{(n+1)!} (0.5)^{n+1}$$

Try $n = 9$:

$$E_9 = \frac{e^{0.5}}{10!} (0.5)^{10} = 4.4369 \times 10^{-10} > 10^{-10}$$

Try $n = 10$:

$$E_{10} = \frac{e^{0.5}}{11!} (0.5)^{11} = 2.0168 \times 10^{-11} < 10^{-10}$$

E_{10} is the first error term less than 10^{-10} , so 10 terms are required.

2. Give an example in base-10 floating point arithmetic (with precision t of your choosing) where

(a)

$$(a + b) + c \neq a + (b + c)$$

(b)

$$(a \times b) \times c \neq a \times (b \times c)$$

Ans: For floating point precision $t=3$, the addition and multiplication roundoff error introduces the inequalities:

a)

$$a = 1.00$$

$$b = 0.0023$$

$$c = 0.0029$$

Left side:

$$(a + b) = fl(1.00 + 0.0023) = fl(1.0023) = 1.00$$

$$(a + b) + c = fl(1.00 + 0.0029) = \mathbf{1.00}$$

Right side:

$$(b + c) = fl(0.0023 + 0.0029) = fl(0.0052) = 0.01$$

$$a + (b + c) = fl(1.00 + 0.01) = \mathbf{1.01}$$

In this example,

$$(a + b) + c = \mathbf{1.00} \neq \mathbf{1.01} = a + (b + c)$$

b)

$$a = 10^3$$

$$b = 0.001$$

$$c = 1.00$$

Left side:

$$(a \times b) = fl(10^3 \times 0.001) = fl(1.00) = 1.00$$

$$(a \times b) \times c = fl(1.00 \times 1.00) = \mathbf{1.00}$$

Right side:

$$(b \times c) = fl(0.001 \times 1.00) = fl(0.001) = 0.00$$

$$a \times (b \times c) = fl(10^3 \times 0.00) = \mathbf{0.00}$$

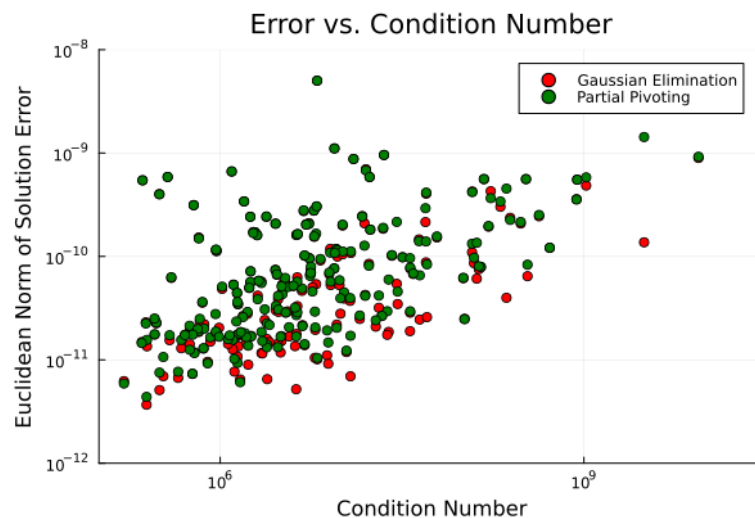
In this example,

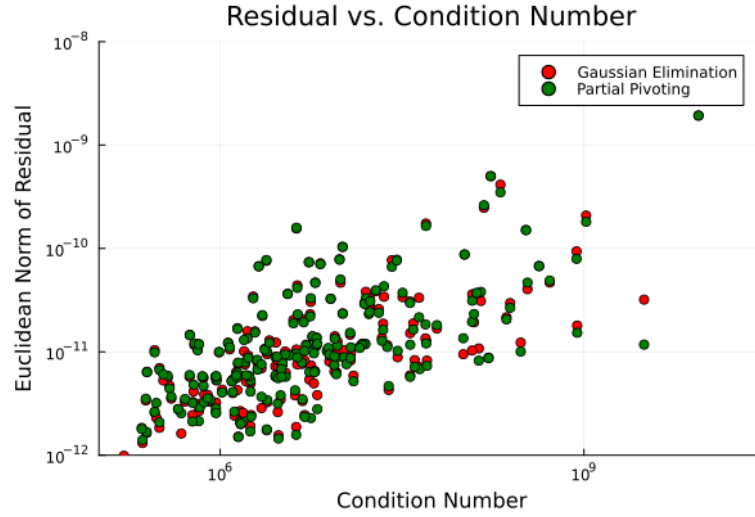
$$(a + b) + c = \mathbf{1.00} \neq \mathbf{0.00} = a + (b + c)$$

3. After implementing the solutions to Problems 4-7, run `plot_error.jl` and examine the four plots that this script creates.

Ans:

- (a) Examine the scatter plots comparing the norm of the residual and the norm of the absolute solution error as a function of the condition number of the system matrix A . Can you see any trends? How do the residual and the absolute solution error differ? How does pivoting affect the performance of Gaussian elimination? Provide a mathematical explanation for what you observe.





These scatter plots comparing the norm of the solution error and the norm of the residual to the condition number show a roughly linear relationship in the log-log space. Condition number on the x-axis represents the sensitivity of the solution to changes in the input data, calculated using the `cond(A)` function in Julia. It indicates how much relative errors or rounding errors can be amplified in the output.

On the first y-axis, the Euclidean Norm of Solution Error represents the distance between the computed solution $\hat{y}$ and the true solution y , or forward error, where

$$e_f = \|\hat{y} - y\|_2$$

The solution error grows according to a power-law trend:

$$e \propto \kappa^m$$

Which appears linear in the log-log space:

$$\log(e) \propto m \log(\kappa)$$

This can be explained by the iterative numerical methods used to solve the linear system. My implementations of Gaussian Elimination without and with partial pivoting (GE and GEPP) introduce small rounding errors on the order of machine epsilon when performing arithmetic operations. When rounding to nearest in the finite precision of our floating point system,

$$fl(x) = x(1 + \epsilon)$$

Where

$$\epsilon \leq u = \epsilon_{mach}/2$$

These errors accumulate and propagate when performing $O(n^3)$ operations (addition, subtraction, multiplication, division) in GE, producing a nonzero forward error. Amplified errors accumulate multiplicatively, causing the absolute error to grow exponentially with the condition number.

On the second y-axis, the residual norm quantifies how well the computed solution satisfies the original equation:

$$\|r\|_2 = \|b - A\hat{x}\|_2$$

After solving the system $Ax = b$ for a computed solution $\hat{x}$, the backward error is:

$$e_{backward} = x - \hat{x}$$

Forward error is amplified by the condition number, where:

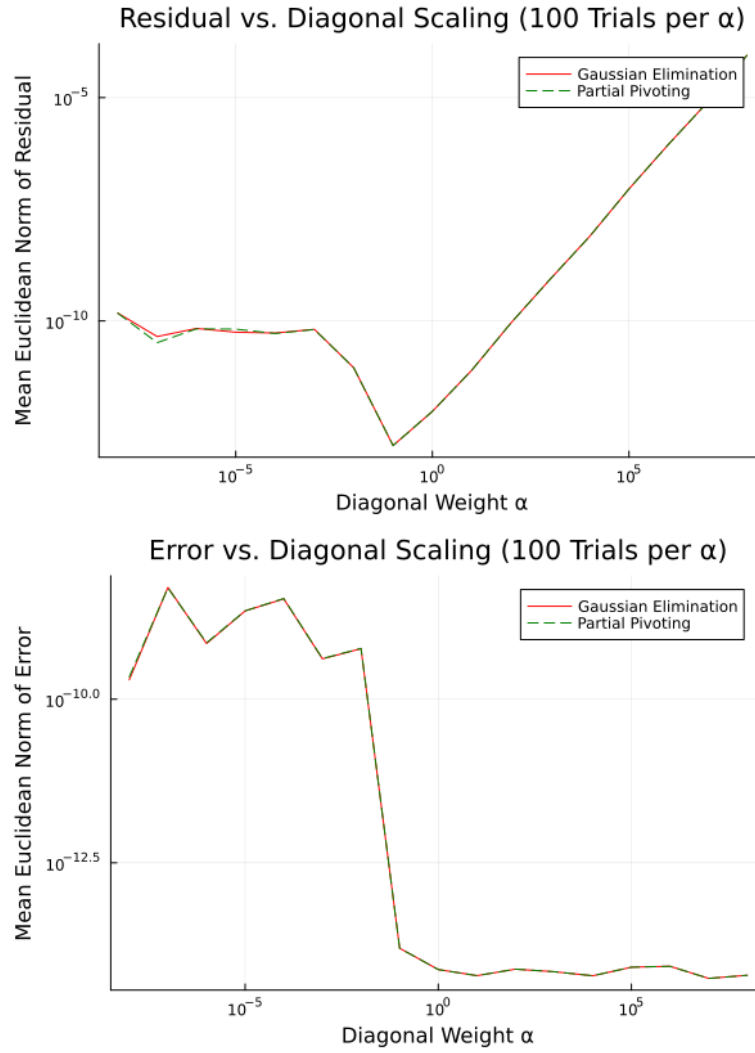
$$e_{forward} \propto \kappa \times e_{backward}$$

$$e_{backward} \propto \frac{e_{forward}}{\kappa}$$

Which explains why the values on the residual norm plot are smaller than the solution error norm plot.

Comparing GE to GEPP, I would expect the vertical spread of GEPP to be smaller, since pivoting should reduce catastrophic outliers compared with no pivoting. It avoids dividing by a very small pivot element $A[k, k]$, which would amplify rounding errors. Without pivoting, if $A[k, k]$ is close to zero, the multiplier $L[i, k] = A[i, k]/A[k, k]$ could blow up in magnitude, causing catastrophic propagation of floating-point error. Pivoting should substantially improve the behaviour by keeping the residuals and errors small, compared to large scatter and catastrophic errors without pivoting. In the graphs produced by my pivoting algorithm, there is no apparent reduction in errors, suggesting that it could be better optimized.

- (b) Examine the plots comparing the mean norm of the residual and the mean norm of the absolute solution error as a function of the diagonal scaling factor $\alpha > 0$ in the modified system matrix $A + \alpha I$. Can you see any trends? How do the residual and the absolute solution error differ? How does pivoting affect the performance of Gaussian elimination? Provide a mathematical explanation for what you observe.



In the plot of α increasing in $A + \alpha I$, the mean residual is moderate for small α , then reaches a minimum before increasing for very large α . In comparison, the mean error is largest for small α , drops at the same value as the residual, and remains small for large α .

The opposite tendencies for very large α can be explained by the effect of adding αI . For very large α , the matrix is dominated by αI which has the effect of pushing small singular values of A away from zero. The condition number $\kappa(A + \alpha I)$ decreases and the system becomes better conditioned. This greatly reduces forward error, which can be seen in the small error values for large α . In contrast, the norm $\|r\|_2 = \|b - (A + \alpha I)\hat{x}\|_2$ grows with α so the residual norm is dominated by this term for large α .

There appears to be no difference in the GE and GEPP lines on both plots,

suggesting that partial pivoting did not affect the results. In theory, for large α pivoting should have little effect because the added diagonal weight prevents the large element growth pivoting is designed to guard against. For α below the order of ϵ_{mach} , diagonal scaling should have minimal effect. The multiplier $L[i, k] = A[i, k]/A[k, k]$ should be unaffected since the precision is below that of our floating point system. The graphs should display the same theoretical trend as in part a), where pivoting decreases catastrophic cancellations to improve the solutions.